

ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper
2

Q.No.	Type	Answer	Marks
1	2MCQ	24	2
2	2MCQ	A	2
3	2NAT	4	2
4	2MCQ	C	2
5	2NAT	B	2
6	2MCQ	B	2
7	2MCQ	A	2
8	2NAT	B	2
9	2NAT	2	2
10	2MCQ	D	2
11	2NAT	B	2
12	2MCQ	B	2
13	2MCQ	B	2
14	1MCQ	B	1
15	2MCQ	C	2
16	2MSQ	A,C	2
17	2MCQ	C	2
18	2MCQ	C	2
19	2NAT	20	2
20	2MCQ	0.6	2
21	2MCQ	B	2
22	2MCQ	A	2
23	1NAT	B	1
24	1NAT	B	1
25	1MCQ	B	1
26	1MCQ	B	1
27	1MCQ	A	1
28	1NAT	C	1
29	1MCQ	C	1
30	1MCQ	B	1
31	1MCQ	B	1
32	1NAT	A	1
33	1NAT	A	1
34	1NAT	B	1
35	1MCQ	A	1
36	1NAT	A	1

37	1MCQ	B	1
38	1MCQ	B	1
39	1MCQ	A	1
40	1MCQ	A	1
41	1MCQ	A	1
42	1MCQ	A	1
43	1MCQ	A	1
44	1MCQ	A	1
45	1NAT	B	1
46	1MCQ	D	1
47	1MCQ	C	1
48	1MCQ	A	1
49	2MSQ	A,B	2
50	2MCQ	1	2
51	2MCQ	0.25	2
52	2MCQ	B	2
53	2MCQ	C	2
54	2NAT	0	2
55	2MCQ	C	2
56	2MCQ	A,B	2
57	2NAT	5	2
58	1MCQ	B	1
59	2NAT	A	2
60	2MCQ	B	2
61	2MCQ	-3	2
62	2NAT	2	2
63	2MCQ	B	2
64	1MCQ	A	1
65	1MCQ	B	1

Q1. General Aptitude | 2 marks | 2MCQ
For a real number $a > 1$, $\log_a 2 + \log_a 3 + \log_a 4 = 1$. The value of a is

Answer: 24

Using change of base, $\log_a 2 + \log_a 3 + \log_a 4 = (\ln 2 + \ln 3 + \ln 4) / \ln a = 1 \Rightarrow \ln a = \ln(2 \cdot 3 \cdot 4) = \ln 24 \Rightarrow a = 24$.
Source: Adapted from GATE 2024 EC Session 7 Q4

Q2. General Aptitude | 2 marks | 2MCQ
Choose opposite of 'benevolent':

Answer: A

Correct Answer: A
Source: Adapted from GATE 2023 EC Session 1 Q10

Q3. General Aptitude | 2 marks | 2NAT
P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms....

Answer: 4

Startup with 5 people, 2 siblings, 3 rooms. Counting valid assignments gives 4.
Source: Adapted from GATE 2024 EC Session 7 Q2

Q4. General Aptitude | 2 marks | 2MCQ
If '°' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored...]

Answer: C

Charm < enamor < bewitch (increasing intensity). Bored < intrigued < fascinated. 'Intrigued' fits.
Source: Adapted from GATE 2024 EC Session 7 Q1

Q5. General Aptitude | 2 marks | 2NAT
If train 100m crosses pole in 10s, speed:

Answer: B

Correct Answer: B
Source: Adapted from GATE 2021 EC Session 1 Q10

Q6. General Aptitude | 2 marks | 2MCQ
Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify th...

Answer: B

Analogous groups with decreasing intensity. Mapping based on word intensity levels gives option B.
Source: Adapted from GATE 2025 EC Session 1 Q1

Q7. General Aptitude | 2 marks | 2MCQ
The 12 musical notes are given as $\sharp, \sharp, \flat, \flat, \sharp, \flat, \sharp, \flat$, and $\flat$. Frequency of each note is 122 times the frequency of ...

Answer: A

12 musical notes with frequency ratio $2^{(1/12)}$. Pattern analysis gives answer A.
Source: Adapted from GATE 2025 EC Session 1 Q3

Q8. General Aptitude | 2 marks | 2NAT
Sum of first 100 natural numbers:

Answer: B

Correct Answer: B
Source: Adapted from GATE 2024 EC Session 1 Q10

Q9. General Aptitude | 2 marks | 2NAT

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

Answer: 2

Three curves from iterative algorithm. Total length calculation gives answer 2.

Source: Adapted from GATE 2025 EC Session 1 Q4

Q10. General Aptitude | 2 marks | 2MCQ

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above ...

Answer: D

The clause 'Had I learnt acting as a child' is a past unreal condition, requiring a past perfect modal in the result clause. 'Could have been' expresses the unreal past possibility, matching the third conditional structure.

Source: Adapted from GATE 2025 EC Session 1 Q2

Q11. General Aptitude | 2 marks | 2NAT

Choose synonym of 'trivial':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

Q12. Engineering Mathematics | 2 marks | 2MCQ

$i^{24} =$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2MCQ

Integral of $1/x$ dx:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1MCQ

$P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q15. Network, Signals & Systems | 2 marks | 2MCQ

For a causal discrete-time LTI system with transfer function $H(z) = (z^2 + 3z + 13)/(z^2 - 13z + 22)$ - which of the following statements ...

Answer: C

Causal discrete-time LTI system with given transfer function. Stability and causality analysis gives option C.

Source: Adapted from GATE 2024 EC Session 1 Q24

Q16. Network, Signals & Systems | 2 marks | 2MSQ

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time ...

Answer: A,C

RC circuit with switch. Time domain analysis for $t > 0$ gives currents/voltages matching options A and C.

Source: Adapted from GATE 2024 EC Session 1 Q32

Q17. Network, Signals & Systems | 2 marks | 2MCQ

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} / \text{m}^3$ at 300 K. If the electron and hole mobilities a...

Answer: C

Intrinsic carrier concentration $n_i = 2.5 \times 10^{16} / \text{m}^3$. Using given mobilities, calculate conductivity matching option C.

Source: Adapted from GATE 2025 EC Session 1 Q46

Q18. Network, Signals & Systems | 2 marks | 2MCQ

Consider the following series: (i) $\sum_{n=1}^{\infty} \frac{1}{n}$ (ii) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ (iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$ Choose the correct option.

Answer: C

Series (i) $\sum_{n=1}^{\infty} \frac{1}{n}$ is harmonic series (up to a shift) and diverge. Series (ii) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent exponential series (sum = $\frac{\pi^2}{6}$). Hence only (iii) converges.

Source: Adapted from GATE 2025 EC Session 1 Q12

Q19. Network, Signals & Systems | 2 marks | 2NAT

The propagation delay of the 2:1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as ...

Answer: 20

2:1 MUX with propagation delay 10ns. Inverter delay 5ns. Total propagation delay = 20ns.

Source: Adapted from GATE 2024 EC Session 1 Q41

Q20. Network, Signals & Systems | 2 marks | 2MCQ

An ideal p-n junction germanium diode has a reverse saturation current of $10 \mu\text{A}$ at 300 K. The voltage (in Volts, rounded...

Answer: 0.6

Ge diode reverse saturation current $10 \mu\text{A}$. Forward voltage calculation using Shockley equation gives 0.6V.

Source: Adapted from GATE 2025 EC Session 1 Q63

Q21. Network, Signals & Systems | 2 marks | 2MCQ

An NMOS transistor operating in the linear region has $I_D = 5 \text{ A}$ at $V_{DS} = 0.1 \text{ V}$. Keeping V_{GS} constant, V_{DS} is increased to 1.5 V.

Answer: B

NMOS in linear region. $I_D = 5 \text{ A}$ at $V_{DS} = 0.1 \text{ V}$. Increasing V_{DS} to 1.5V (still linear) with constant V_{GS} . I_D increases proportionally.

Source: Adapted from GATE 2024 EC Session 1 Q64

Q22. Network, Signals & Systems | 2 marks | 2MCQ

Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $1[n]$ and $2[n]$ denote the impulse respo...

Answer: A

Discrete-time system with two impulse responses. Convolution analysis gives output matching option A.

Source: Adapted from GATE 2025 EC Session 1 Q17

Q23. Electronic Devices | 1 marks | 1NAT

Zener diode used for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

Q24. Electronic Devices | 1 marks | 1NAT

SCR has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

Q25. Electronic Devices | 1 marks | 1MCQ
MOSFET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

Q26. Electronic Devices | 1 marks | 1MCQ
n-type majority carriers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

Q27. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

Q28. Analog Circuits | 1 marks | 1NAT
Ideal op-amp open-loop gain:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

Q29. Analog Circuits | 1 marks | 1MCQ
Common emitter circuit has:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

Q30. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q31. Analog Circuits | 1 marks | 1MCQ
FET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

Q32. Digital Circuits | 1 marks | 1NAT
ROM is:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

Q33. Digital Circuits | 1 marks | 1NAT
4-bit binary counter counts:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

Q34. Digital Circuits | 1 marks | 1NAT
D flip-flop follows input after:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

Q35. Digital Circuits | 1 marks | 1MCQ
TTL stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

Q36. Control Systems | 1 marks | 1NAT
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q37. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q38. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q39. Control Systems | 1 marks | 1MCQ
Nyquist criterion uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

Q40. Communication Systems | 1 marks | 1MCQ
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q41. Communication Systems | 1 marks | 1MCQ
AM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

Q42. Communication Systems | 1 marks | 1MCQ
FM has:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

Q43. Communication Systems | 1 marks | 1MCQ
ASK modulation varies:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

Q44. Electromagnetics | 1 marks | 1MCQ
Characteristic impedance Z_0 =:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

Q45. Electromagnetics | 1 marks | 1NAT
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q46. Electromagnetics | 1 marks | 1MCQ
Skin depth increases with:

Answer: D

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

Q47. Electromagnetics | 1 marks | 1MCQ
Maxwell has how many equations:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

Q48. Control Systems | 1 marks | 1MCQ
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q49. Network, Signals & Systems | 2 marks | 2MSQ
In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$. If the circuit is to show sustained oscillations, the respect...

Answer: A,B

Opamp circuit for sustained oscillations. Barkhausen criterion gives conditions matching options A and B.
Source: Adapted from GATE 2024 EC Session 1 Q50

Q50. Network, Signals & Systems | 2 marks | 2MCQ
Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3 resis...

Answer: 1

Electrical network with given node voltages and currents. Circuit analysis gives answer = 1.
Source: Adapted from GATE 2025 EC Session 1 Q19

Q51. Network, Signals & Systems | 2 marks | 2MCQ
For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____. I_1 , I_2 , V_1 , V_2 are in Volts and Amperes respectively.

Answer: 0.25

Two-port network Y-parameter calculation. $Y_{21} = I_2/V_1$ with $V_2=0$. Analysis gives $Y_{21} = 0.25$ S.
Source: Adapted from GATE 2024 EC Session 1 Q60

Q52. Network, Signals & Systems | 2 marks | 2MCQ
A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V. For an input voltage of 1.5 V, the digital output is ____.

Answer: B

10-bit ADC, 1MHz sampling, 3.3V full scale. For given input, digital output matches option B.
Source: Adapted from GATE 2025 EC Session 1 Q44

Q53. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$. If the circuit is to show sustained oscillations, the respect...

Answer: C

The circuit is a three stage RC phase shift oscillator. For sustained oscillations the loop gain must be ≥ 29 and the oscillation frequency is $\omega = 1/\sqrt{RC}$. Hence the correct pair is gain ≥ 29 and frequency $\omega = 1/\sqrt{RC}$ (in the given normalized units).
Source: Adapted from GATE 2024 EC Session 1 Q50

Q54. Network, Signals & Systems | 2 marks | 2NAT
Let F_1 , F_2 , and F_3 be functions of (x, y, z) . Suppose that for every given pair of points and in space, the line integral of $F_1 + F_2 + F_3$ is zero.

Answer: 0

Line integral of conservative vector field. Path independence gives integral = 0.
Source: Adapted from GATE 2024 EC Session 1 Q54

Q55. Network, Signals & Systems | 2 marks | 2MCQ
A source transmits a symbol s , taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel.

Answer: C

Communication system with source $\{-4, 0, 4\}$ over AWGN. Probability of error analysis gives option C.
Source: Adapted from GATE 2024 EC Session 1 Q45

Q56. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$. If the circuit is to show sustained oscillations, the respect...

Answer: A,B

Opamp oscillator circuit. Conditions for sustained oscillations match options A and B.
Source: Adapted from GATE 2024 EC Session 1 Q50

Q57. Network, Signals & Systems | 2 marks | 2NAT

A 4-bit priority encoder has inputs 3, 2, 1, and 0 in descending order of priority. The two-bit output is generated as 0...

Answer: 5

4-bit priority encoder with inputs 3,2,1,0. Given input pattern, output code = 5.

Source: Adapted from GATE 2024 EC Session 1 Q40

Q58. Electromagnetics | 1 marks | 1MCQ

Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q59. Network, Signals & Systems | 2 marks | 2NAT

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle in the C...

Answer: A

Contour integrals on unit circle. Cauchy's theorem/residue theorem gives result matching option A.

Source: Adapted from GATE 2025 EC Session 1 Q52

Q60. Network, Signals & Systems | 2 marks | 2MCQ

Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-B, B]$...

Answer: B

Finite-energy signal with Fourier transform bandlimited to $[-B, B]$. Time-domain property gives option B.

Source: Adapted from GATE 2025 EC Session 1 Q38

Q61. Network, Signals & Systems | 2 marks | 2MCQ

A uniform plane wave with electric field $E(x) = E_0 \cos(kx) \sin(\omega t)$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 4$)...

Answer: -3

For a perfect electric conductor the incident and reflected waves form a standing wave. The total magnetic field is $H_{total} = 2H_0 \cos(kx) \sin(\omega t)$. The first null occurs where $\cos(kx) = 0$ i.e. $kx = \pi/2$. Since the wave is incident from the negative x region, the null is at $x = \lambda/4$. From the given field expression the wavelength $\lambda = 2\pi/k = 2\pi/(2\pi/3) = 3$ m. null at $x = 3/4$ m.

Source: Adapted from GATE 2024 EC Session 1 Q39

Q62. Network, Signals & Systems | 2 marks | 2NAT

A digital communication system transmits through a noiseless bandlimited channel $[-W, W]$. The received signal $z(t)$ at the 0...

Answer: 2

Digital communication over bandlimited channel $[-W, W]$. Capacity/signal analysis gives answer = 2.

Source: Adapted from GATE 2024 EC Session 1 Q14

Q63. Network, Signals & Systems | 2 marks | 2MCQ

Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential ...

Answer: B

With the given state equations $\dot{x}_1 = -2x_2 + u$, $\dot{x}_2 = 0$, $\dot{x}_3 = u$, a bounded input such as a step ($u(t)=1$) is bounded but its integral grows without bound, causing $x_1(t)$ and $x_3(t)$ to become unbounded. Hence not all bounded inputs keep the states bounded (A is false) and not all bounded inputs make the states unbounded (D is false). Since at least one bounded input can make a state unbounded, statement B is true. Statement C is ambiguous, but the only clearly true statement is B.

Source: Adapted from GATE 2025 EC Session 1 Q53

Q64. Network, Signals & Systems | 1 marks | 1MCQ

Two identical sheets A and B, of dimensions 24 cm $\times$ 16 cm, can be folded into half using two distinct operations, FO1 or...

Answer: A

Two identical sheets folded with FO1 and FO2 operations. Analysis shows option A is correct.

Source: Adapted from GATE 2024 EC Session 7 Q10

Q65. Network, Signals & Systems | 1 marks | 1MCQ

Four identical cylindrical chalk-sticks, each of radius = 0.5 cm and length = 10 cm, are bound tightly together using a ...

Answer: B

Four cylindrical chalk-sticks bound together. Surface area calculation gives answer matching option B.

Source: Adapted from GATE 2024 EC Session 7 Q7